

Second Semester Math Assignment

Course: Mathematics-II (Course No: 2.1)

Answer the following questions based on the prescribed syllabus.

Unit I: Introduction to Differential Calculus

1. Evaluate the limit: $\lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 5x}$.
2. Evaluate the indeterminate form using L'Hôpital's Rule: $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$.
3. Find the limit: $\lim_{x \rightarrow \infty} \left(1 + \frac{2}{x}\right)^x$.
4. Determine the right-hand limit: $\lim_{x \rightarrow 0^+} x \ln x$.
5. Evaluate the limit representing the indeterminate form $\infty - \infty$: $\lim_{x \rightarrow \pi/2} (\sec x - \tan x)$.
6. Discuss the continuity of the piecewise function $f(x) = \frac{|x|}{x}$ at $x = 0$.
7. Find the value of the constant k such that the function $f(x) = \frac{\sin 2x}{x}$ for $x \neq 0$, and $f(0) = k$, is continuous at $x = 0$.
8. Examine the differentiability of the absolute value function $f(x) = |x - 2|$ at the point $x = 2$.
9. Differentiate the function $y = x^{\sin x}$ with respect to x using logarithmic differentiation.
10. Find $\frac{dy}{dx}$ implicitly if $x^3 + y^3 = 3axy$, where a is a constant.
11. Differentiate the composite function $y = \ln(\sec x + \tan x)$ with respect to x .
12. Find the derivative of the inverse trigonometric function $y = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$.
13. If $x = a(\cos \theta + \theta \sin \theta)$ and $y = a(\sin \theta - \theta \cos \theta)$, determine the parametric derivative $\frac{dy}{dx}$.
14. Compute the first-order partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for $f(x, y) = e^{x^2+y^2}$.
15. Verify that the mixed partial derivatives are equal ($\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$) for the function $z = x^2 \sin y + y^2 \cos x$.
16. If $u = \ln(x^3 + y^3 + z^3 - 3xyz)$, prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$.
17. Evaluate the total derivative $\frac{du}{dt}$ using the chain rule if $u = x^2 + y^2$, given $x = e^t$ and $y = e^{-t}$.
18. Find the partial derivative $\frac{\partial z}{\partial x}$ implicitly if the variables are related by $x^2 + y^2 + z^2 - 2xyz = 1$.
19. Evaluate the limit of the indeterminate form 1^∞ : $\lim_{x \rightarrow 0} (\cos x)^{1/x^2}$.
20. Check the continuity and differentiability of the function defined by $f(x) = x \sin(1/x)$ for $x \neq 0$ and $f(0) = 0$ at the origin $x = 0$.

Unit II: Expansion of Functions

21. State Rolle's Theorem and verify it for the polynomial $f(x) = x^2 - 4x + 3$ on the interval $[1, 3]$.
22. Verify Rolle's Theorem for the trigonometric function $f(x) = \sin x + \cos x$ on the closed interval $[0, 2\pi]$.
23. Apply Lagrange's Mean Value Theorem to the function $f(x) = x^3 - x$ on $[-1, 1]$ and locate the value(s) of c .
24. Verify Lagrange's Mean Value Theorem for the logarithmic function $f(x) = \ln x$ on the interval $[1, e]$.
25. State Cauchy's Mean Value Theorem clearly.
26. Verify Cauchy's Mean Value Theorem for the pair of functions $f(x) = x^2$ and $g(x) = x^3$ on the interval $[1, 2]$.
27. Expand $\sin x$ in ascending powers of x using Maclaurin's Theorem.
28. Obtain the Maclaurin series expansion for the exponential function e^x .
29. Use Maclaurin's series to expand $\ln(1 + x)$ up to the term containing x^4 .
30. Find the Taylor series expansion of $f(x) = \cos x$ centered about the point $x = \pi/4$.
31. Expand the polynomial $x^3 - 2x^2 + 3x - 5$ in positive powers of $(x - 2)$ using Taylor's Theorem.
32. Use the first three terms of a suitable Taylor's series to approximate $\sqrt{26}$ to three decimal places.
33. Define a homogeneous function of degree n and state Euler's Theorem for two variables.
34. Verify Euler's Theorem for the homogeneous polynomial function $u(x, y) = x^2 + xy + y^2$.
35. If $u = \sin^{-1} \left(\frac{x^2 + y^2}{x + y} \right)$, mathematically prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \tan u$.
36. If $u = \tan^{-1} \left(\frac{x^3 + y^3}{x - y} \right)$, determine the exact value of $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$.
37. If $z = f(x/y)$, demonstrate that $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 0$.
38. Apply Maclaurin's theorem to find the binomial expansion of $(1 + x)^n$ where n is any real number.
39. Verify Rolle's theorem for the product function $f(x) = e^x \sin x$ on the interval $[0, \pi]$.
40. Determine the first three non-zero terms of the Maclaurin series expansion for the function $f(x) = \tan x$.

Unit III: Maxima and Minima

41. Find the local maxima and local minima of the cubic function $f(x) = 2x^3 - 9x^2 + 12x - 5$.
42. Determine all the critical points of the function $f(x) = x^4 - 4x^3 + 4x^2$.
43. Find the absolute maximum and minimum values of $f(x) = \sin x + \cos x$ on the closed interval $[0, \pi]$.

44. Locate the coordinates of the points of inflection for the curve given by $y = x^4 - 6x^2 + 8x + 10$.
45. A rectangular box with an open top is to be formed from a square piece of cardboard 12 cm on a side by cutting squares from the corners and folding. Find the dimensions that yield the maximum volume.
46. Find the absolute maximum and minimum values of the function $f(x) = x^3 - 3x^2 + 1$ on the interval $[-1/2, 4]$.
47. Determine the dimensions of a rectangle of maximum possible area that can be inscribed inside a circle of fixed radius R .
48. Calculate the maximum profit possible if a company's profit function is modelled by $P(x) = -2x^2 + 400x - 5000$.
49. Find the critical points of the two-variable function $f(x, y) = x^2 + xy + y^2 - 3x - 3y$.
50. Examine the polynomial function $f(x, y) = x^3 + y^3 - 3xy$ for extreme values and classify them.
51. Find the local maxima, minima, and saddle points of the surface defined by $f(x, y) = x^4 + y^4 - 4xy + 1$.
52. Determine the nature of the critical points for the function $f(x, y) = xy(1 - x - y)$.
53. Find the minimum distance from the origin $(0, 0, 0)$ to the plane defined by $x - 2y + 2z = 6$.
54. Investigate the local extrema of the trigonometric function $f(x, y) = \sin x + \sin y + \sin(x + y)$ within the region $0 \leq x, y \leq \pi$.
55. Use the Second Partials Test to classify the critical points of $f(x, y) = 3x^2y + y^3 - 3x^2 - 3y^2 + 2$.
56. Find the extreme values of the objective function $f(x, y) = x^2y$ subject to the constraint $x^2 + 2y^2 = 6$.
57. Find the dimensions of an open-top rectangular box of maximum capacity whose total surface area is exactly 432 cm^2 .
58. Discuss the maxima and minima behavior of the function $f(x, y) = x^3y^2(1 - x - y)$.
59. Formulate the minimization function to determine the point on the surface $z = x^2 - y^2$ that is closest to the spatial point $(1, 1, 0)$.
60. Identify the coordinates of any saddle points present on the hyperbolic paraboloid surface $z = x^2 - y^2$.

Unit IV: Integral Calculus

61. Evaluate the indefinite integral using integration by parts: $\int xe^x dx$.
62. Derive the standard integral formula for $\int \frac{dx}{x^2 - a^2}$.
63. Evaluate the rational integral $\int \frac{3x+1}{x^2-5x+6} dx$ using the method of partial fractions.
64. Evaluate the definite trigonometric integral $\int_0^{\pi/2} \cos^2 x dx$.
65. Evaluate the integral $\int_0^1 x\sqrt{1-x^2} dx$ using an appropriate algebraic substitution.
66. Use properties of odd/even functions in definite integrals to evaluate $\int_{-\pi/2}^{\pi/2} \sin^7 x \cos^4 x dx$.

67. Evaluate the definite integral $\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx$.
68. Derive the standard reduction formula for $I_n = \int \sin^n x dx$.
69. Using the established reduction formula, compute the numerical value of $\int_0^{\pi/2} \sin^6 x dx$.
70. Obtain the recursive reduction formula for the indefinite integral $\int \tan^n x dx$.
71. Find the exact area enclosed between the quadratic curve $y = x^2$ and the linear line $y = x$.
72. Calculate the geometric area of the region bounded by the parabola $y^2 = 4ax$ and its corresponding latus rectum.
73. Find the arc length of the planar curve $y = \frac{2}{3}x^{3/2}$ evaluated from $x = 0$ to $x = 3$.
74. Formulate an integral to determine the perimeter of the circle $x^2 + y^2 = r^2$ and evaluate it.
75. Find the volume of the solid generated by completely revolving the region bounded by $y = x^2$ and $y = 4$ about the y-axis.
76. Calculate the volume of a sphere of radius r by revolving the upper semicircle $y = \sqrt{r^2 - x^2}$ about the x-axis.
77. Evaluate the iterated double integral $\int_0^1 \int_0^2 (x^2 + y^2) dy dx$.
78. Evaluate the double integral $\iint_R xy dA$, where the region R is bounded by $y = x^2$ and $y = x$.
79. Evaluate the straightforward triple integral $\int_0^1 \int_0^z \int_0^{y+z} dx dy dz$.
80. Evaluate the double integral given in polar coordinates: $\int_0^{\pi/2} \int_0^{\cos \theta} r^2 \sin \theta dr d\theta$.

Unit V: Transform Calculus

81. Find the standard Laplace transform of the polynomial function $f(t) = t^3$.
82. Determine the Laplace transform of the exponential function $L\{e^{3t}\}$.
83. Evaluate the linear Laplace transform of the combination $L\{\sin 4t + \cos 4t\}$.
84. State the First Shifting Theorem and use it to calculate $L\{e^{2t} \cos 3t\}$.
85. Find the Laplace transform of the product function $f(t) = t^2 e^{-3t}$.
86. Evaluate $L\{t \sin 2t\}$ by applying the "multiplication by t " property of Laplace transforms.
87. Find the Laplace transform $L\left\{\frac{\sin t}{t}\right\}$ using the "division by t " property.
88. Determine the Laplace transform of the piecewise step function: $f(t) = 1$ for $0 < t < 2$ and $f(t) = 0$ for $t > 2$.
89. Evaluate the Laplace transform of the expanded binomial $(t + 2)^2$.
90. Find the Laplace transforms for the hyperbolic functions $L\{\cosh at\}$ and $L\{\sinh at\}$.
91. Find the inverse Laplace transform $L^{-1}\left\{\frac{1}{s^2 - 16}\right\}$.
92. Evaluate the inverse Laplace transform $L^{-1}\left\{\frac{s}{s^2 + 25}\right\}$.

93. Determine $L^{-1} \left\{ \frac{1}{(s-3)^4} \right\}$ by employing the inverse First Shifting Theorem.
94. Find the time-domain function given by $L^{-1} \left\{ \frac{s-2}{(s-2)^2+9} \right\}$.
95. Evaluate $L^{-1} \left\{ \frac{1}{s(s+2)} \right\}$ by breaking the expression into partial fractions.
96. Find the inverse Laplace transform of the rational expression $\frac{3s+1}{s^2-s-6}$.
97. State the Convolution Theorem as applied to inverse Laplace transforms.
98. Use the Convolution Theorem to evaluate the inverse transform $L^{-1} \left\{ \frac{1}{(s^2+a^2)^2} \right\}$.
99. Evaluate the inverse transform involving a logarithm $L^{-1} \left\{ \ln \left(\frac{s+1}{s-1} \right) \right\}$ using the derivative property.
100. Find the inverse Laplace transform $L^{-1} \left\{ \frac{s^2}{(s^2+4)(s^2+9)} \right\}$.